Impact of a Rule-based Recommendation System on Female Students' Awareness of How to Search for Suitable Professional Certificates

D.J. Robinson, K.J. Alvarez

Abstract

This study examines the impact of an intelligent, rule-based recommendation system designed to address a key gap in literature; while most AI-based advising tools focus on course selection, few guide learners toward professional certifications that build career readiness. A quasi-experimental pre–post design was conducted with sixty-three female undergraduate students at a Saudi university, where limited awareness of certification pathways has been noted. Data were collected through a validated electronic questionnaire, and analyses were performed using Python for accuracy and transparency. Grounded in the Technology Acceptance Model (TAM), the study interprets awareness gains through students' engagement, motivation, and perceived usefulness of the system. Results from the paired-samples t-Test showed significant improvements in students' understanding of effective search strategies and their awareness of officially endorsed certificates and associated costs. Although the effect size was small (Cohen's d = 0.28), the findings indicate that even short exposure to an AI-driven tool can enhance professional awareness. Practically, the study highlights the potential of such systems to support academic advising and align higher education outcomes with labor market needs.